In [43]:
```
import pandas as pd
```

In [44]:
```
df = pd.read_csv(r'/Users/abhiramireddygali/Downloads/data.csv')
```

In [45]:
```
df.shape
```

Out[45]:
```
(195, 5)
```

In [46]:
```
df.columns
```

Out[46]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [47]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   CountryName    195 non-null    object
 1   CountryCode    195 non-null    object
 2   BirthRate      195 non-null    float64
 3   InternetUsers  195 non-null    float64
 4   IncomeGroup    195 non-null    object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

In [48]:
```
df.describe()
```

Out[48]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [49]:
```
len(df)
```

Out[49]:
```
195
```

In [50]:
```
type(df)
```

Out[50]: pandas.core.frame.DataFrame

In [51]:
```
len(df.columns)
```

Out[51]: 5

In [52]:
```
df.head()
```

Out[52]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [53]:
```
df.tail()
```

Out[53]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [54]:
```
df.tail(2)
```

Out[54]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [55]:
```
df[::-1]
```

Out[55]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

In [56]:

```
df[:5]
```

Out[56]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [57]:

```
df[6:]
```

Out [57]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income
10	Azerbaijan	AZE	18.300	58.7000	Upper middle income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0000	Lower middle income
191	South Africa	ZAF	20.850	46.5000	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2000	Low income
193	Zambia	ZMB	40.471	15.4000	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5000	Low income

189 rows × 5 columns

In [58]:

```
df[0:200:10]
```

Out [58]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
30	Canada	CAN	10.900	85.800000	High income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
60	Gabon	GAB	30.555	9.200000	Upper middle income
70	Greenland	GRL	14.500	65.800000	High income
80	India	IND	20.291	15.100000	Lower middle income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
110	Moldova	MDA	12.141	45.000000	Lower middle income
120	Mozambique	MOZ	39.705	5.400000	Low income
130	Netherlands	NLD	10.200	93.956400	High income
140	Poland	POL	9.600	62.849200	High income
150	Sudan	SDN	33.477	22.700000	Lower middle income
160	Suriname	SUR	18.455	37.400000	Upper middle income
170	Tajikistan	TJK	30.792	16.000000	Lower middle income
180	Uruguay	URY	14.374	57.690000	High income
190	Yemen, Rep.	YEM	32.947	20.000000	Lower middle income

In [59]:

```
df[0:200:50]
```

Out[59]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
150	Sudan	SDN	33.477	22.700000	Lower middle income

In [60]:
```
df.describe()
```

Out[60]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [61]:
```
df.describe().transpose()
```

Out[61]:

	count	mean	std	min	25%	50%	75%	
BirthRate	195.0	21.469928	10.605467	7.9	12.1205	19.68	29.7595	4
InternetUsers	195.0	42.076471	29.030788	0.9	14.5200	41.00	66.2250	9

In [62]:
```
df.describe().T
```

Out[62]:

	count	mean	std	min	25%	50%	75%	
BirthRate	195.0	21.469928	10.605467	7.9	12.1205	19.68	29.7595	4
InternetUsers	195.0	42.076471	29.030788	0.9	14.5200	41.00	66.2250	9

In [63]:
```
df.columns
```

Out[63]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [64]:
```
df.columns = ['a','b','c','d','e']
```

In [65]:
```
df.head()
```

Out[65]:

	a	b	c	d	e
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [66]:
```
df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetU
          'IncomeGroup']
```

In [67]:
```
df.columns
```

Out[67]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [68]:
```
df.head(1)
```

Out[68]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income

In [69]:
```
df.isnull()
```

Out[69]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [70]:
```
df.isnull().sum()
```

Out[70]:
```
CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```

In [71]:
```
df.dtypes
```

Out[71]:
```
CountryName       object
CountryCode       object
BirthRate        float64
InternetUsers    float64
IncomeGroup       object
dtype: object
```

In [72]:
```
df.columns
```

Out[72]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [76]:
```
df_categorical = df[['CountryName', 'CountryCode','IncomeGroup']]
```

In [77]:
```
df_categorical.head()
```

Out[77]:

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income

In [75]:
```
type(df)
```

Out[75]:
```
pandas.core.frame.DataFrame
```

In [79]:
```
df.describe()
```

Out[79]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [80]:
```
df_categorical.describe()
```

Out[80]:

	CountryName	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

In [83]:
```
df_num = df[['BirthRate', 'InternetUsers']]
```

In [84]:
```
df_num
```

Out[84]:

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
...	...	...
190	32.947	20.0
191	20.850	46.5
192	42.394	2.2
193	40.471	15.4
194	35.715	18.5

195 rows × 2 columns

In [85]:
```
df_num.describe().T
```

Out[85]:

	count	mean	std	min	25%	50%	75%
BirthRate	195.0	21.469928	10.605467	7.9	12.1205	19.68	29.7595
InternetUsers	195.0	42.076471	29.030788	0.9	14.5200	41.00	66.2250

In [86]:
```
df['IncomeGroup']
```

Out[86]:
```
0                High income
1                 Low income
2        Upper middle income
3        Upper middle income
4                High income
                 ...
190      Lower middle income
191      Upper middle income
192               Low income
193      Lower middle income
194               Low income
Name: IncomeGroup, Length: 195, dtype: object
```

In [87]:
```
df[4:7][['CountryName','BirthRate']]
```

Out[87]:

	CountryName	BirthRate
4	United Arab Emirates	11.044
5	Argentina	17.716
6	Armenia	13.308

In [88]:
```
df.head(1)
```

Out[88]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income

In [89]:
```
df.BirthRate * df.InternetUsers
```

Out[89]:
```
0      808.2516
1      207.9927
2      878.3135
3      736.5644
4      971.8720
         ...
190    658.9400
191    969.5250
192     93.2668
193    623.2534
194    660.7275
Length: 195, dtype: float64
```

In [90]:
```
df['BirthRate'] * df['InternetUsers']
```

Out[90]:
```
0      808.2516
1      207.9927
2      878.3135
3      736.5644
4      971.8720
         ...
190    658.9400
191    969.5250
192     93.2668
193    623.2534
194    660.7275
Length: 195, dtype: float64
```

In [105…
```
df['myCalc'] = df['BirthRate'] * df['InternetUsers']
```

In [106…
```
df.columns
```

Out[106…
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup', 'myCalc'],
      dtype='object')
```

In [107…
```
df
```

Out[107…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	
0	Aruba	ABW	10.244	78.9	High income	8
1	Afghanistan	AFG	35.253	5.9	Low income	2
2	Angola	AGO	45.985	19.1	Upper middle income	8
3	Albania	ALB	12.877	57.2	Upper middle income	7
4	United Arab Emirates	ARE	11.044	88.0	High income	9
...	...	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income	6
191	South Africa	ZAF	20.850	46.5	Upper middle income	9
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income	
193	Zambia	ZMB	40.471	15.4	Lower middle income	6
194	Zimbabwe	ZWE	35.715	18.5	Low income	6

195 rows × 6 columns

In [108…
```
df.isnull().sum()
```

Out[108…
```
CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
myCalc           0
dtype: int64
```

In [109…
```
len(df.columns)
```

Out[109…
```
6
```

In [110…
```
df.columns
```

Out[110…
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup', 'myCalc'],
      dtype='object')
```

In [111…
```
df.drop('myCalc' ,axis = 1) # axis =1 (column) & axis =0 (rows)
```

Out[111…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [112…
```
df.columns
```

Out[112…
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup', 'myCalc'],
      dtype='object')
```

In [113…
```
df.columns[3:4]
```

Out[113…
```
Index(['InternetUsers'], dtype='object')
```

In [114…
```
df.InternetUsers < 2
```

Out[114…
```
0      False
1      False
2      False
3      False
4      False
       ...
190    False
191    False
192    False
193    False
194    False
Name: InternetUsers, Length: 195, dtype: bool
```

In [115…
```
df
```

Out[115…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	
0	Aruba	ABW	10.244	78.9	High income	8
1	Afghanistan	AFG	35.253	5.9	Low income	2
2	Angola	AGO	45.985	19.1	Upper middle income	8
3	Albania	ALB	12.877	57.2	Upper middle income	7:
4	United Arab Emirates	ARE	11.044	88.0	High income	9
...	...	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income	6!
191	South Africa	ZAF	20.850	46.5	Upper middle income	9
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income	!
193	Zambia	ZMB	40.471	15.4	Lower middle income	6
194	Zimbabwe	ZWE	35.715	18.5	Low income	6

195 rows × 6 columns

In [120…
```
Filter = df.InternetUsers < 2
```

In [121…
```
Filter
```

Out[121…
```
0      False
1      False
2      False
3      False
4      False
       ...
190    False
191    False
192    False
193    False
194    False
Name: InternetUsers, Length: 195, dtype: bool
```

In [122…
```
df[Filter]
```

Out[122…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	n
11	Burundi	BDI	44.151	1.3	Low income	5
52	Eritrea	ERI	34.800	0.9	Low income	3
55	Ethiopia	ETH	32.925	1.9	Low income	6
64	Guinea	GIN	37.337	1.6	Low income	5
117	Myanmar	MMR	18.119	1.6	Lower middle income	2
127	Niger	NER	49.661	1.7	Low income	8
154	Sierra Leone	SLE	36.729	1.7	Low income	6
156	Somalia	SOM	43.891	1.5	Low income	6
172	Timor-Leste	TLS	35.755	1.1	Lower middle income	3

In [125…
```
len(df[Filter])
```

Out[125… 9

In [126…
```
Filter2 = df.BirthRate > 40
Filter2
```

Out[126…

```
0      False
1      False
2       True
3      False
4      False
       ...
190    False
191    False
192     True
193     True
194    False
Name: BirthRate, Length: 195, dtype: bool
```

In [127…

```
df[Filter2]
```

Out[127…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	
2	Angola	AGO	45.985	19.1	Upper middle income	8
11	Burundi	BDI	44.151	1.3	Low income	
14	Burkina Faso	BFA	40.551	9.1	Low income	3
65	Gambia, The	GMB	42.525	14.0	Low income	5
115	Mali	MLI	44.138	3.5	Low income	1
127	Niger	NER	49.661	1.7	Low income	
128	Nigeria	NGA	40.045	38.0	Lower middle income	15
156	Somalia	SOM	43.891	1.5	Low income	
167	Chad	TCD	45.745	2.3	Low income	1
178	Uganda	UGA	43.474	16.2	Low income	7
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income	
193	Zambia	ZMB	40.471	15.4	Lower middle income	6

In [128…

```
len(df[Filter2])
```

Out[128… 12

In [129…

```
Filter & Filter2
```

Out[129…
```
0      False
1      False
2      False
3      False
4      False
       ...
190    False
191    False
192    False
193    False
194    False
Length: 195, dtype: bool
```

In [130…
```
df[Filter & Filter2]
```

Out[130…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	n
11	Burundi	BDI	44.151	1.3	Low income	5
127	Niger	NER	49.661	1.7	Low income	8
156	Somalia	SOM	43.891	1.5	Low income	6

In [131…
```
df[df.IncomeGroup =='Low income']
```

Out[131…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	
1	Afghanistan	AFG	35.253	5.90	Low income	2
11	Burundi	BDI	44.151	1.30	Low income	
13	Benin	BEN	36.440	4.90	Low income	1
14	Burkina Faso	BFA	40.551	9.10	Low income	3
29	Central African Republic	CAF	34.076	3.50	Low income	1
38	Comoros	COM	34.326	6.50	Low income	2
52	Eritrea	ERI	34.800	0.90	Low income	
55	Ethiopia	ETH	32.925	1.90	Low income	
64	Guinea	GIN	37.337	1.60	Low income	
65	Gambia, The	GMB	42.525	14.00	Low income	5
66	Guinea-Bissau	GNB	37.503	3.10	Low income	1
77	Haiti	HTI	25.345	10.60	Low income	2
93	Cambodia	KHM	24.462	6.80	Low income	1
99	Liberia	LBR	35.521	3.20	Low income	1
111	Madagascar	MDG	34.686	3.00	Low income	1
115	Mali	MLI	44.138	3.50	Low income	1

120	Mozambique	MOZ	39.705	5.40	Low income
123	Malawi	MWI	39.459	5.05	Low income
127	Niger	NER	49.661	1.70	Low income
132	Nepal	NPL	20.923	13.30	Low income
148	Rwanda	RWA	32.689	9.00	Low income
154	Sierra Leone	SLE	36.729	1.70	Low income
156	Somalia	SOM	43.891	1.50	Low income
158	South Sudan	SSD	37.126	14.10	Low income
167	Chad	TCD	45.745	2.30	Low income
168	Togo	TGO	36.080	4.50	Low income
177	Tanzania	TZA	39.518	4.40	Low income
178	Uganda	UGA	43.474	16.20	Low income
192	Congo, Dem. Rep.	COD	42.394	2.20	Low income
194	Zimbabwe	ZWE	35.715	18.50	Low income

In [132…

```
df[df.IncomeGroup == 'High income']
```

Out[132…

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
4	United Arab Emirates	ARE	11.044	88.00	High income
5	Argentina	ARG	17.716	59.90	High income
7	Antigua and Barbuda	ATG	16.447	63.40	High income
8	Australia	AUS	13.200	83.00	High income
...	...	...	...	...	...
174	Trinidad and Tobago	TTO	14.590	63.80	High income
180	Uruguay	URY	14.374	57.69	High income
181	United States	USA	12.500	84.20	High income
184	Venezuela, RB	VEN	19.842	54.90	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.30	High income

67 rows × 6 columns

In [133…
```
df.IncomeGroup.unique()
```

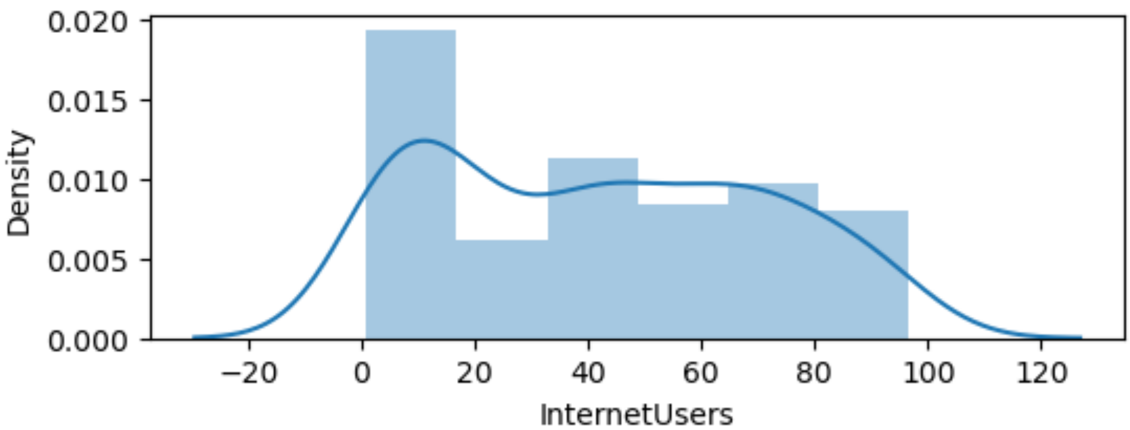
Out[133…
```
array(['High income', 'Low income', 'Upper middle income',
       'Lower middle income'], dtype=object)
```

In [134…
```
df.IncomeGroup.nunique()
```

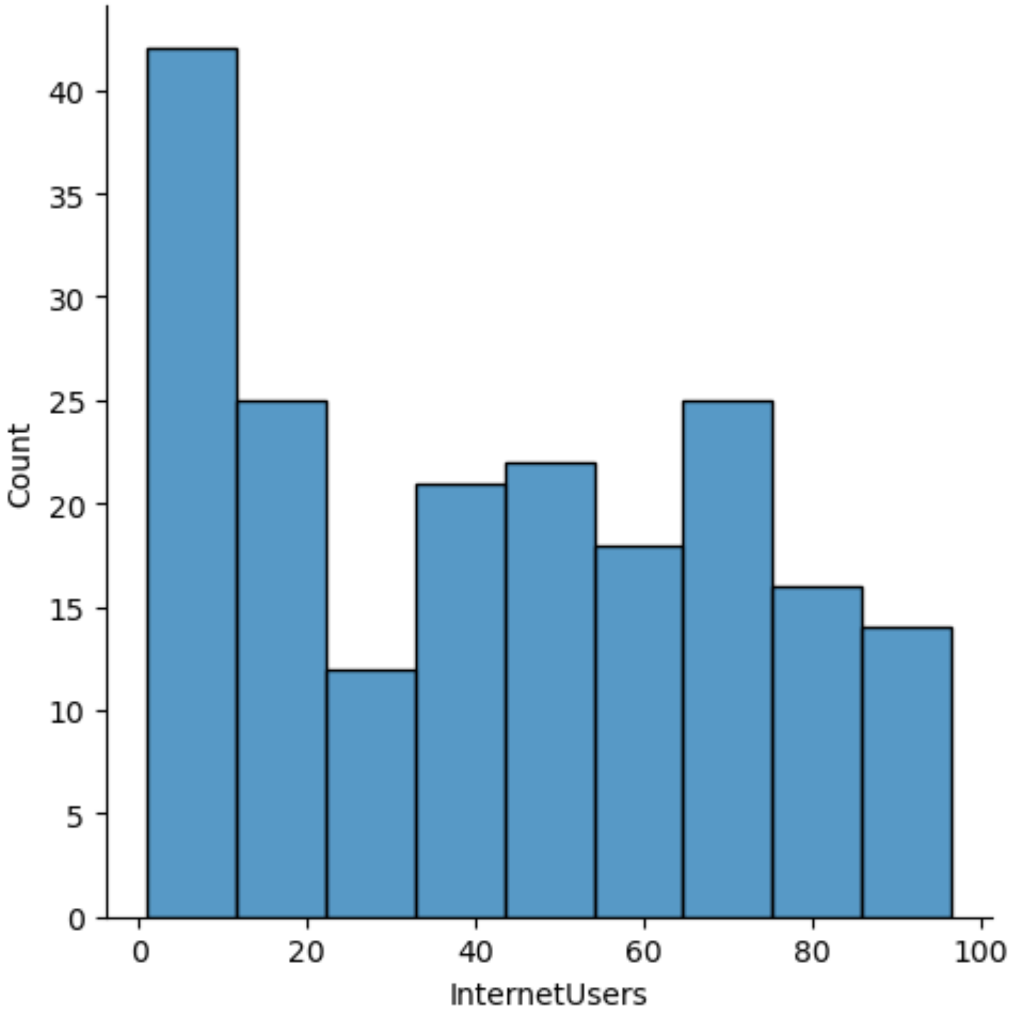
Out[134…
```
4
```

In [136…
```
import matplotlib.pyplot as plt # visulation
import seaborn as sns # distribution visualization

%matplotlib inline
plt.rcParams['figure.figsize'] = 6,2

import warnings
warnings.filterwarnings('ignore')
```

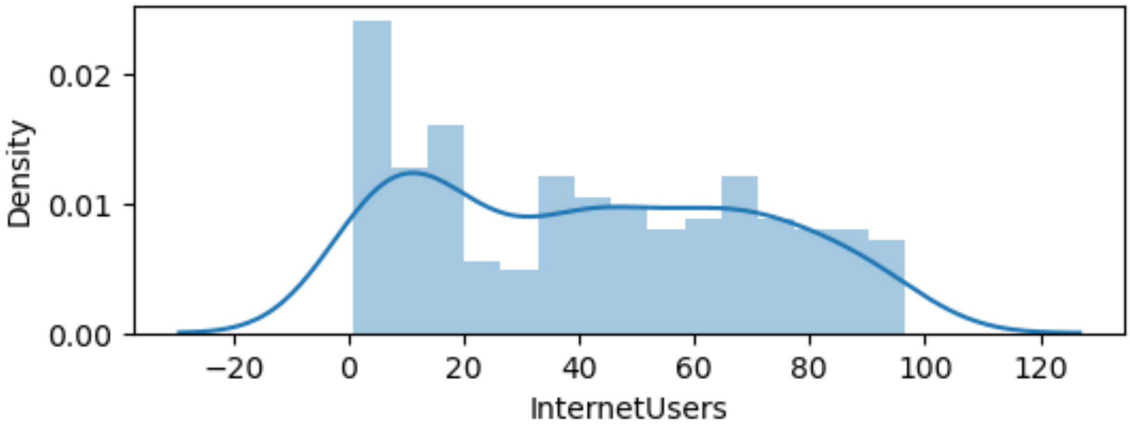
In [137…
```
vis1 = sns.distplot(df['InternetUsers']) # univariate analysis using
```



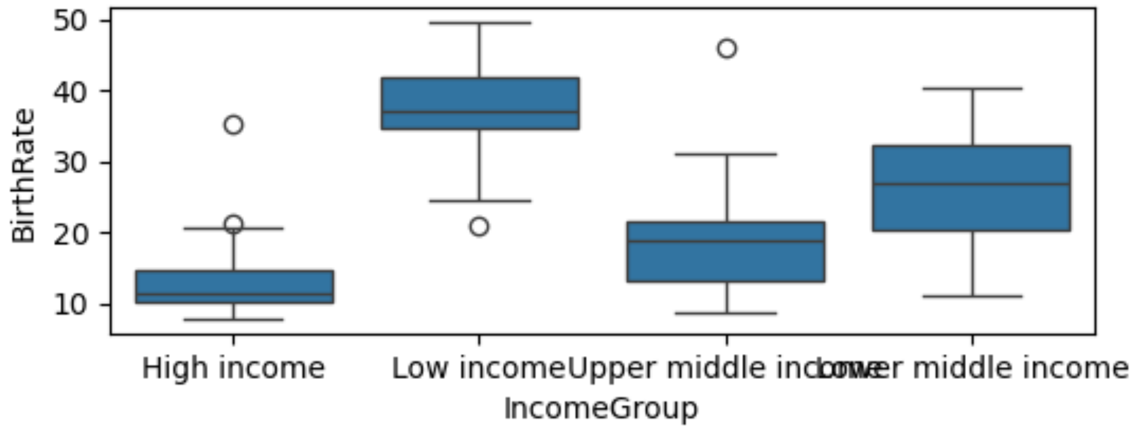
In [138…
```
vis2 = sns.displot(df['InternetUsers'])
```



In [141…
```
vis3 = sns.distplot(df['InternetUsers'],bins = 15)
```

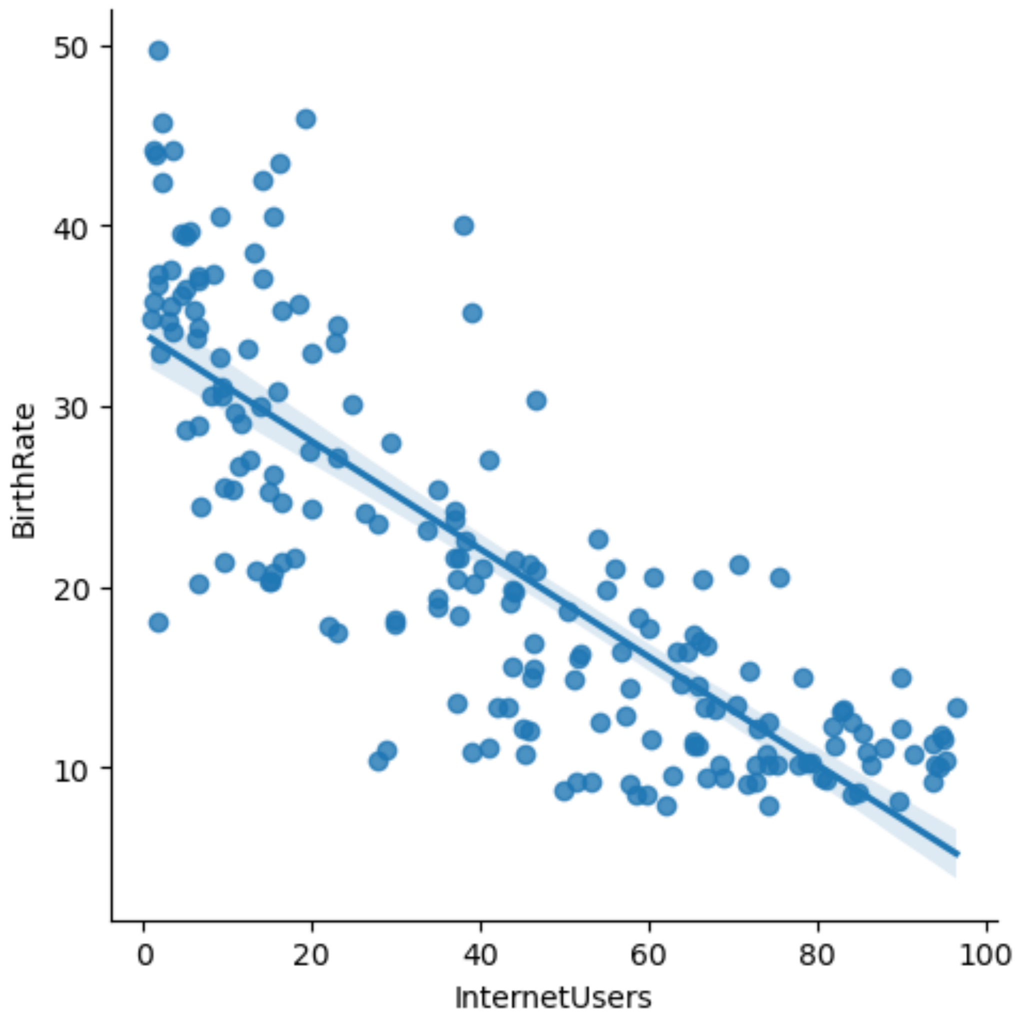


In [143…
```
vis4 =sns.boxplot(data =df, x='IncomeGroup',y='BirthRate') # bi-var
```



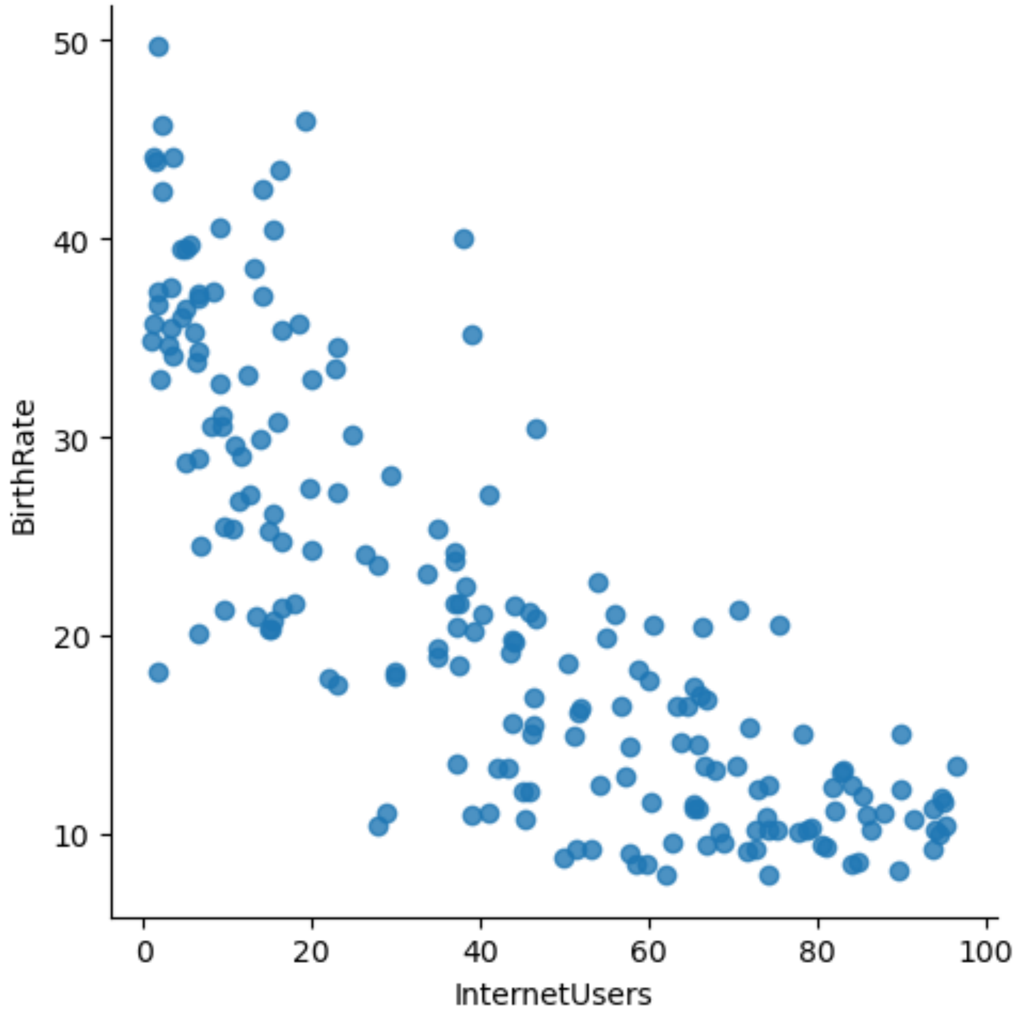
In [145…

```
vis5 =sns.lmplot(data =df, x='InternetUsers',y='BirthRate')
```



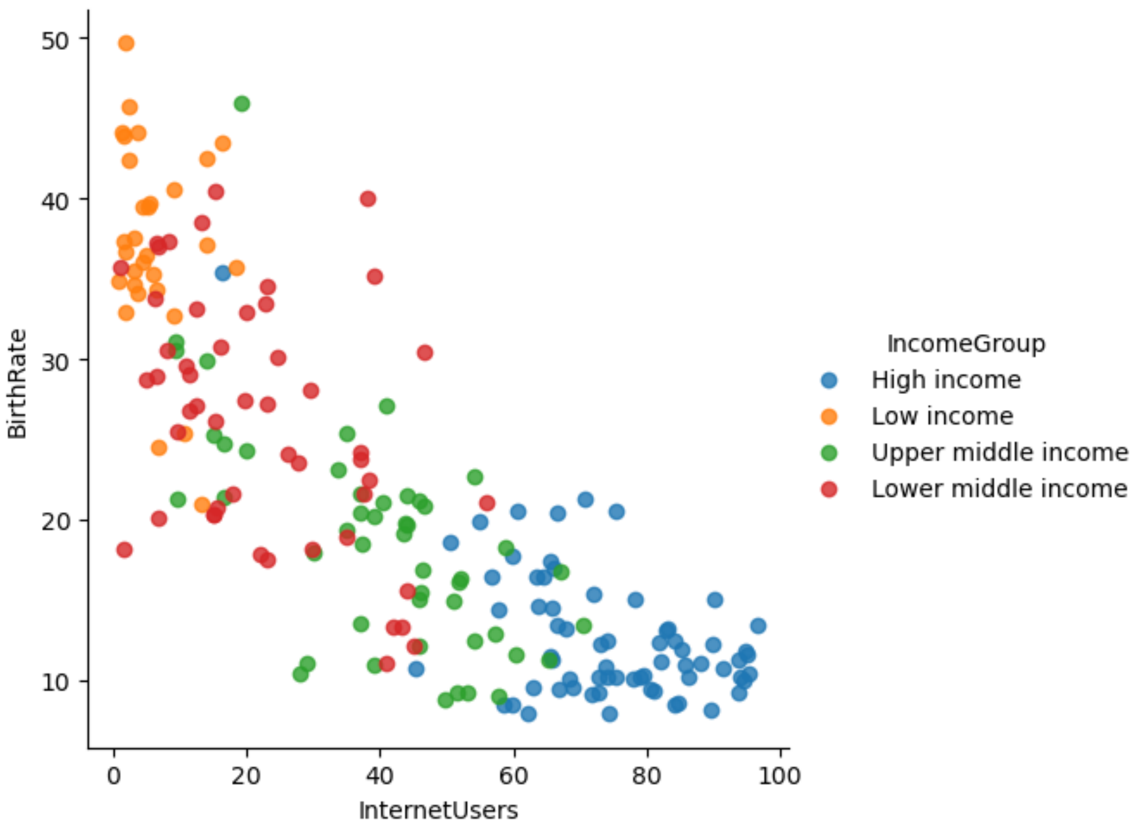
In [146…

```
vis5 =sns.lmplot(data =df, x='InternetUsers',y='BirthRate',fit_reg=
```



In [148…

```
vis5 =sns.lmplot(data =df, x='InternetUsers',y='BirthRate',fit_reg=
```



In [149…

```
vis5 =sns.lmplot(data =df, x='InternetUsers',y='BirthRate',fit_reg='
```